

WCN UNIT - 2

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1.0 An Introduction to Second-Generation (2G) Networks

1.1 The Digital Revolution: Core Advancements over 1G

The advent of second-generation (2G) wireless networks in the early 1990s marked a paradigm shift from the analog-based first-generation (1G) systems. This transition to digital communication technology was not merely an incremental upgrade but a fundamental revolution that laid the groundwork for the modern mobile era. The switch to digital brought a host of transformative benefits that addressed the significant limitations of 1G systems.¹

Key advancements of 2G over 1G include:

- **Enhanced Speech Quality:** Digital encoding and error-correction techniques provided clearer, more consistent voice quality and improved immunity to noise and interference compared to the analog systems.¹
- **Increased Capacity and Spectrum Efficiency:** Digital technologies, particularly Time Division Multiple Access (TDMA) and Code Division Multiple Access (CDMA), allowed multiple users to share the same frequency channel, significantly increasing the number of subscribers a network could support within a given spectrum allocation.¹
- **Introduction of Data Services:** For the first time, digital pathways enabled the transmission of data, opening the door for services like the Short Message Service (SMS), which became a cornerstone of mobile communication.¹
- **Enhanced Security:** The digital nature of 2G allowed for the implementation of robust authentication and over-the-air encryption, making calls significantly more secure and protecting user privacy from casual eavesdropping, a major vulnerability of 1G analog networks.¹
- **Improved System Features:** Digital call processing enabled more sophisticated features such as caller ID, call forwarding, and conference calling, which were difficult or impossible to implement in analog systems.¹
- **Standardization and Roaming:** 2G saw the development of comprehensive standards like the Global System for Mobile (GSM), which aimed to solve the fragmentation problem of 1G. This standardization enabled international roaming, allowing subscribers to use their mobile devices in different countries.¹

1.2 Foundational 2G Access Technologies: An Overview

The efficiency and capacity of 2G networks were built upon two primary multiple access schemes: TDMA and CDMA. The choice between these technologies represented two distinct philosophies for managing the finite radio spectrum resource.

- **Time Division Multiple Access (TDMA):** This technology operates on the principle of time-sharing. A single radio frequency channel is divided into multiple time slots, and each user is allocated a specific slot to transmit or receive information. It is a "scheduling" approach, dictating *when* each user can access the channel. TDMA systems require precise synchronization between the mobile device and the base station to ensure that transmissions do not overlap and interfere with one another. To prevent interference from minor timing errors, guard intervals (short periods of silence) are inserted between time slots.¹ TDMA was an evolutionary step, applying the principles of time-division multiplexing, already common in wired telephone networks, to the wireless domain.
- **Code Division Multiple Access (CDMA):** This technology takes a more radical approach based on spread-spectrum techniques originally developed for military use. In a CDMA system, all users transmit simultaneously over the same wide frequency band. It is an "interference management" approach. To distinguish between users, each subscriber is assigned a unique pseudo-random code. The user's data is multiplied by this code, which spreads the signal across the wide bandwidth. The receiver, knowing the specific code, can then "de-spread" the signal and recover the original data, while the signals from all other users appear as background noise. This method provides high security and is inherently resistant to interference.¹

The divergence between these two technologies led to the parallel development of competing 2G standards, with GSM adopting TDMA and standards like IS-95 championing CDMA.

2.0 Global System for Mobile (GSM): Architecture and Services

2.1 Introduction and Design Goals

The Global System for Mobile Communications (GSM) is the most widely adopted 2G digital cellular standard globally. It was developed by the European Telecommunications Standards Institute (ETSI) to address the shortcomings and fragmentation of the first-generation analog systems across Europe. The primary goal was to create a unified, open standard that would provide a common set of compatible services and capabilities to all mobile users, enabling seamless international roaming.¹

The fundamental design requirements of the GSM standard were:

- **Service Portability:** To allow subscribers to access communication services in all participating countries.¹
- **Quality of Service (QoS):** To ensure that the voice quality was at least as good as the preceding analog systems.¹
- **Security:** To provide secure over-the-air transmission through robust authentication and data encryption mechanisms.¹
- **High Spectrum Efficiency:** To utilize the allocated radio frequency bands as efficiently as possible to maximize network capacity.¹
- **Cost-Effectiveness:** To design system parameters with a focus on reducing the overall cost of the network infrastructure and subscriber equipment.¹

2.2 GSM System Services

Following the guidelines of the Integrated Services Digital Network (ISDN), GSM services are categorized into three distinct groups, offering a range of capabilities beyond simple voice calls.¹

1. **Bearer (Data) Services:** These services are responsible for the transmission of data between user-network interfaces and correspond to the lower three layers of the OSI model. They provide the fundamental capability to transport data bits.
 - They support data rates from 300 bps to 9.6 kbps.
 - They offer both synchronous and asynchronous data transmission.
 - Services can be *transparent*, providing a fixed-delay connection with no error correction, or *non-transparent*, which uses radio link protocols to provide error correction and flow control for higher reliability.¹
2. **Teleservices (Telephone Services):** These are user-to-user services that provide the

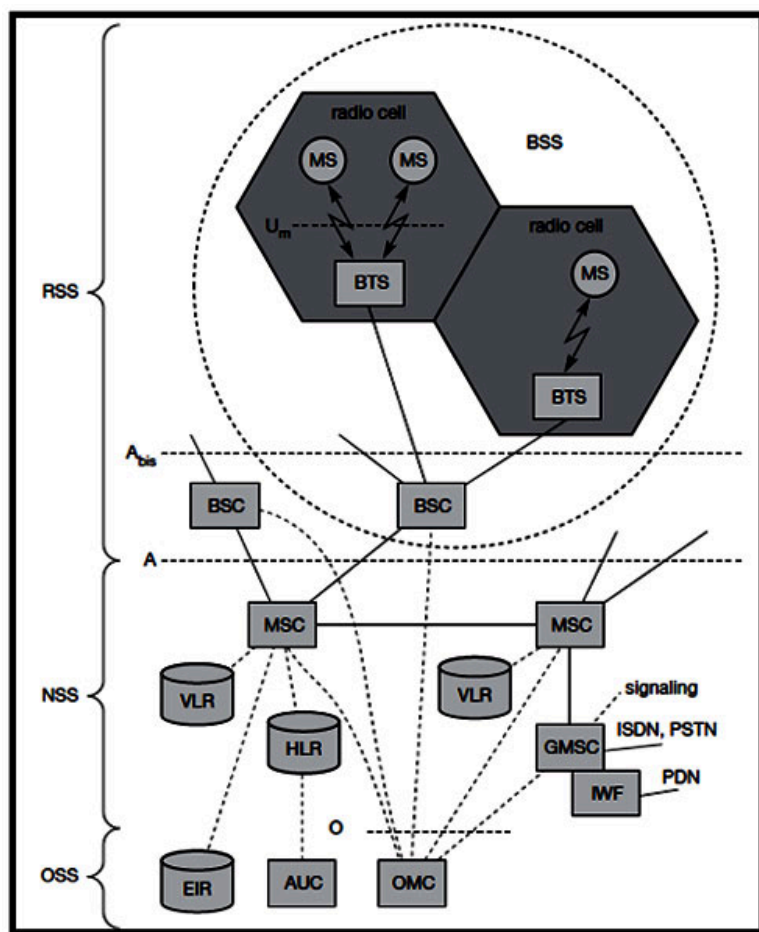
complete capability for communication, including the terminal equipment functions.

- **Standard Telephony:** The primary service for full-duplex voice communication.
- **Emergency Calling:** Provides a standardized way to contact emergency services (e.g., dialing 112).
- **Messaging Services:** Includes Short Message Service (SMS), Multimedia Messaging Service (MMS), and facsimile (fax) services.¹

3. **Supplementary Services:** These are modifications or additions to the basic teleservices, providing more advanced features to the user.

- **Call Identification:** Services like Caller ID.
- **Call Redirection:** Includes Call Forwarding (e.g., on busy, no answer) and Call Diversion.
- **Call Waiting and Call Hold:** Allows users to manage multiple calls simultaneously.
- **Multi-party Communication:** Enables conference calls.¹

2.3 GSM System Architecture [FIG]



The GSM network architecture is logically structured into three major interconnected subsystems. This modular design is a key strength, as it decouples the radio-dependent components from the core switching network and management functions. This separation allows for independent evolution and upgrades of different parts of the network, such as introducing new radio technologies without replacing the entire core infrastructure.¹

2.3.1 The Three Major Subsystems

1. **Base Station Subsystem (BSS):** Also known as the radio subsystem, the BSS is responsible for all functions related to the radio transmission path between the mobile station (MS) and the core network. It manages the radio interface, including channel allocation, handoffs, and signal quality control.¹
2. **Network and Switching Subsystem (NSS):** The NSS is the core of the GSM network, responsible for managing calls and subscriber data. It performs the switching functions of the system, routes calls between mobile users and to external networks like the Public Switched Telephone Network (PSTN) or ISDN, and manages mobility.¹
3. **Operation Support Subsystem (OSS):** The OSS provides the necessary functions for the operation and maintenance of the GSM network. It allows network operators to monitor, diagnose, troubleshoot, and manage all aspects of the system, including billing, traffic monitoring, and hardware maintenance.¹

2.3.2 Mobile Station (MS) and Subscriber Identity Module (SIM) ``

The Mobile Station (MS) is the device used by the subscriber to access GSM services. It consists of two distinct components:

- **Mobile Equipment (ME):** This is the physical handset itself, identified by a unique International Mobile Equipment Identity (IMEI) number. The ME is responsible for radio transmission and reception, speech coding, and other device-level functions.¹
- **Subscriber Identity Module (SIM):** The SIM is a small, removable smart card that securely stores all subscriber-related information. This includes the International Mobile Subscriber Identity (IMSI), which uniquely identifies the user, as well as secret keys used for authentication and encryption. The separation of the subscriber identity (SIM) from the equipment (ME) is a cornerstone of GSM, enabling users to easily switch handsets while retaining their phone number and service profile.¹

2.3.3 The Base Station Subsystem (BSS)

The BSS acts as the bridge between the mobile stations and the NSS. It is composed of two

main functional entities ¹:

- **Base Transceiver Station (BTS):** A BTS contains all the radio equipment—transceivers, antennas, amplifiers—necessary to provide radio coverage for a single cell. Its primary function is to manage the radio link with the mobile stations over the air interface (Um interface). A single BTS can handle several radio channels simultaneously.¹
- **Base Station Controller (BSC):** The BSC is the "brain" of the BSS, managing the resources for one or more BTSs. Its key responsibilities include:
 - **Radio Resource Management:** Assigning radio frequencies and time slots to mobile stations.
 - **Handoff Management:** Controlling handoffs between BTSs under its control (intra-BSC handoff).
 - **Paging:** Initiating the paging of a mobile station for an incoming call.The BSC connects to the BTSs via the Abis interface and to the core network's MSC via the A interface.¹

2.3.4 The Network and Switching Subsystem (NSS)

The NSS is the heart of the GSM network, performing the core switching and mobility management functions. Its key components are ¹:

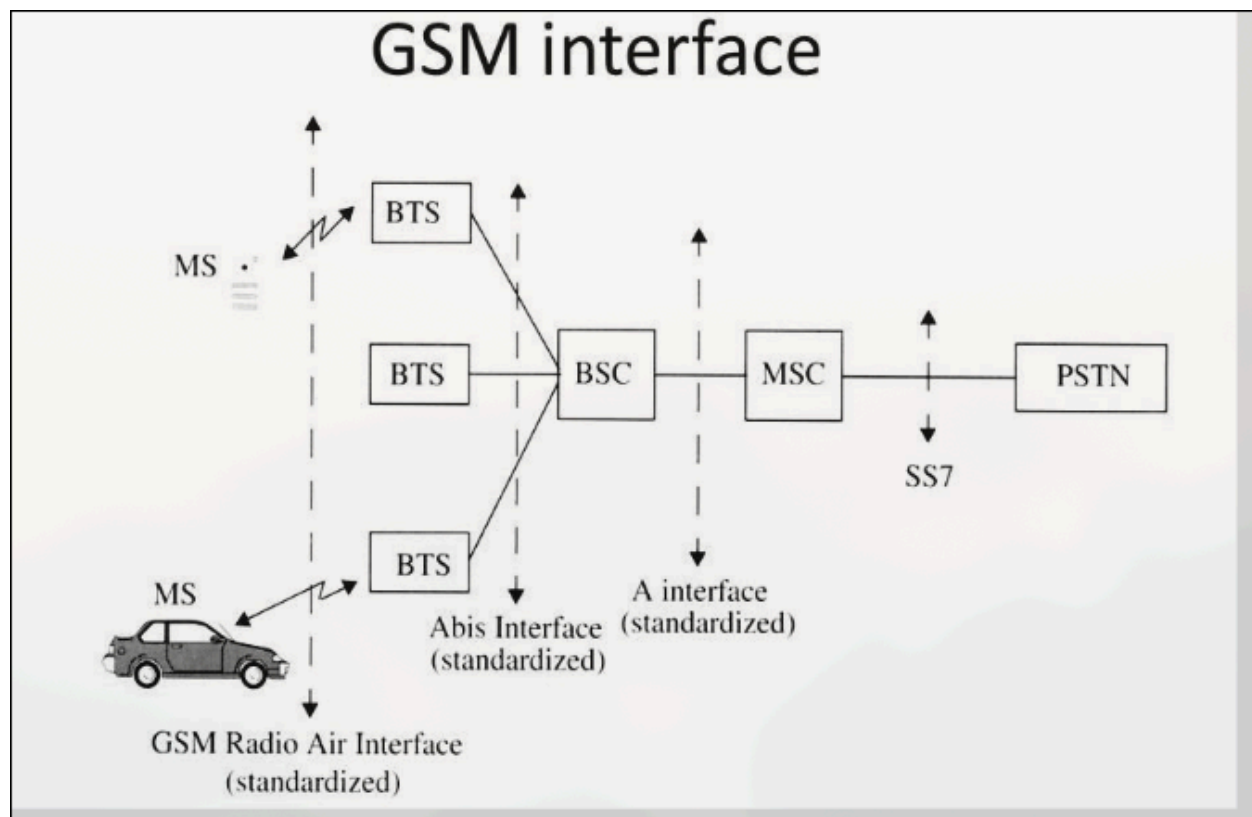
- **Mobile Switching Center (MSC):** The MSC is the central component of the NSS. It acts as a specialized telephone exchange, performing all the switching and signaling functions for mobile subscribers located in the geographical area it serves. It routes calls to and from other MSCs, the BSS, and external networks.¹
- **Home Location Register (HLR):** The HLR is a centralized, permanent database that stores all administrative information for every subscriber registered with the network operator. For each subscriber, the HLR contains their IMSI, service subscriptions (e.g., call forwarding), and, critically, their current location information (i.e., the address of the Visitor Location Register where the subscriber is currently located).¹
- **Visitor Location Register (VLR):** The VLR is a temporary database that is typically co-located with an MSC. It stores a copy of the subscriber profile for all users who are currently roaming within its service area. When a subscriber enters a new MSC area, the VLR queries the HLR for their information and stores it locally. This allows the MSC to access subscriber data for call setup and service handling without having to query the centralized HLR every time, reducing signaling traffic.¹
- **Gateway MSC (GMSC):** The GMSC is a specialized MSC that acts as the interface between the mobile network and external networks like the PSTN. When a call is placed to a GSM subscriber from an external network, it is first routed to the GMSC. The GMSC's primary role is to query the HLR to determine the current location of the called subscriber and then route the call to the appropriate "visited" MSC.¹

2.3.5 The Operation Support Subsystem (OSS)

The OSS contains the functional entities that support the day-to-day management of the network¹:

- **Authentication Center (AuC):** The AuC is a highly secure database associated with the HLR. It stores the secret authentication key (Ki) for each subscriber. Its function is to generate the security parameters (challenge-response triplets) used by the network to authenticate a subscriber and establish encryption keys for the radio link, thus protecting against cloning and fraud.¹
- **Equipment Identity Register (EIR):** The EIR is a database that stores the IMEI numbers for all valid mobile equipment on the network. It maintains lists of valid (white list), monitored (grey list), and stolen or unauthorized (black list) handsets. This allows the network to block services to a specific device if it has been reported stolen.¹
- **Operation and Maintenance Center (OMC):** The OMC is the centralized hub from which network operators monitor and manage the entire GSM system. It provides functionalities for traffic monitoring, network performance analysis, alarm handling, and software updates for network elements like BSCs and MSCs.¹

2.4 Key GSM Interfaces [FIG]



The standardized interfaces between the different subsystems and entities are crucial for ensuring interoperability between equipment from different manufacturers.¹

- **Um (Air Interface):** This is the radio interface between the Mobile Station (MS) and the Base Transceiver Station (BTS). It is the most complex interface in GSM, defining the modulation, channel structure, and signaling protocols for over-the-air communication.¹
- **Abis Interface:** This internal BSS interface connects a Base Transceiver Station (BTS) to its controlling Base Station Controller (BSC). It carries traffic channels as well as control and maintenance data for the BTS.¹
- **A Interface:** This interface connects the Base Station Subsystem (BSS) to the Network and Switching Subsystem (NSS), specifically linking the BSC to the MSC. It uses the Signaling System 7 (SS7) protocol suite to carry signaling messages for call control, mobility management, and radio resource management.¹

3.0 GSM Radio Interface and Signal Processing

3.1 Radio Link Aspects

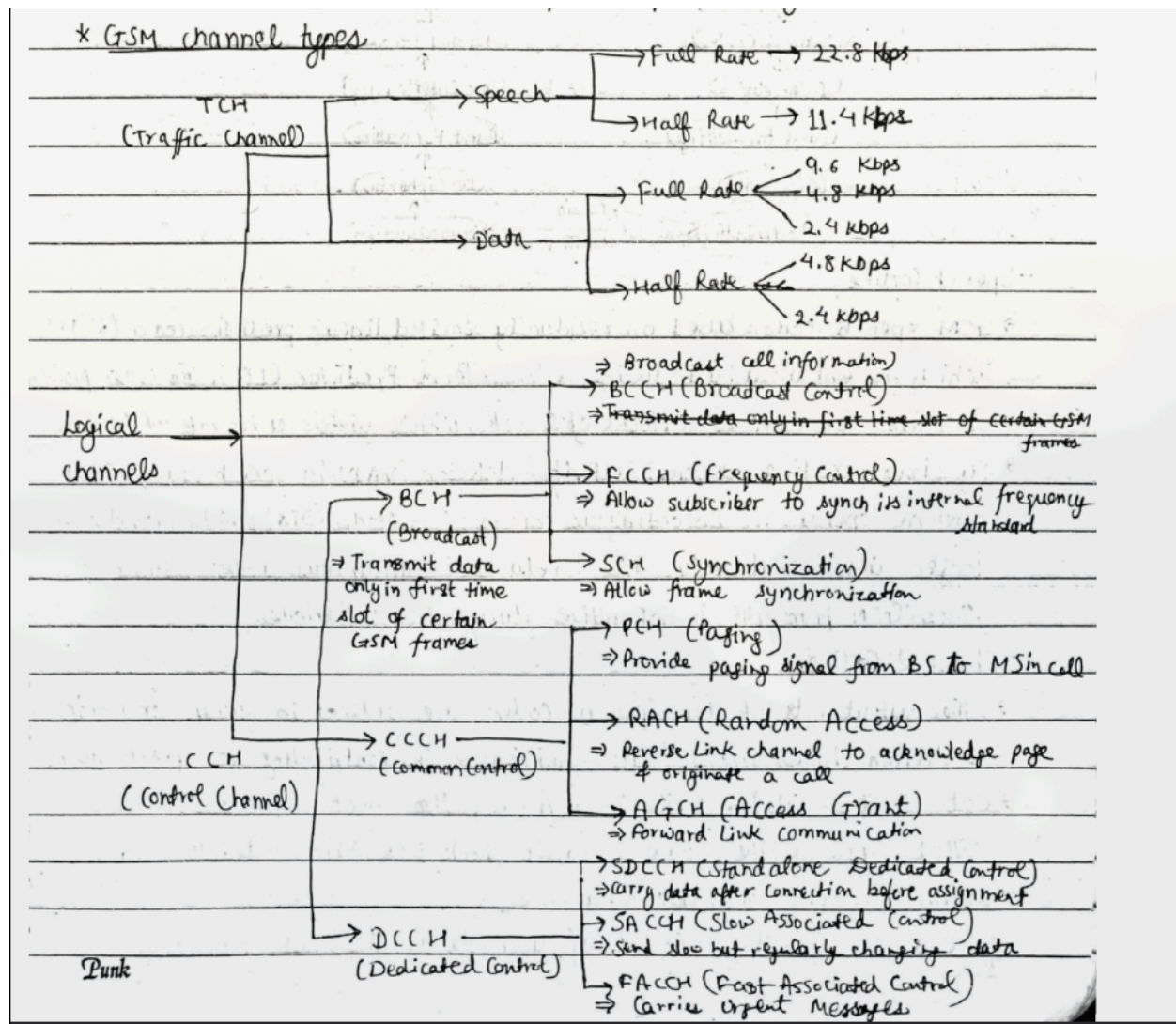
GSM operates using a combination of Frequency Division Duplex (FDD) and a hybrid of Frequency Division Multiple Access (FDMA) and Time Division Multiple Access (TDMA) to manage the radio spectrum.

- **Frequency Bands:** The original GSM-900 standard uses the 890-915 MHz band for the uplink (mobile to base) and the 935-960 MHz band for the downlink (base to mobile). The uplink and downlink frequencies for a given connection are always separated by 45 MHz.¹
- **FDMA Component:** The total 25 MHz bandwidth in each direction is divided into 124 radio frequency channels, each with a bandwidth of 200 kHz. A guard band of 100 kHz is left at each end of the spectrum.¹
- **TDMA Component:** Each of these 200 kHz radio channels is further subdivided in the time domain into 8 time slots. Each time slot lasts approximately 577 microseconds (μ s). A user is assigned one time slot on an uplink frequency and one time slot on a downlink frequency for a full-duplex call. This structure allows up to 8 users to share a single radio channel simultaneously.¹

3.2 GSM Logical Channels [FIG]

While the physical resource is a time slot on a specific frequency, this resource is used to carry different types of information. GSM organizes this information flow using a system of logical channels, which are multiplexed onto the physical channels. These logical channels are

broadly classified into Traffic Channels and Control Channels.¹



3.2.1 Traffic Channels (TCH)

Traffic Channels are used to carry the actual user data, which can be either digitized speech or circuit-switched data.¹

- **Full-Rate TCH (TCH/F):** Carries speech coded at 13 kbps or data at up to 9.6 kbps, occupying one time slot per TDMA frame. The gross data rate is 22.8 kbps after channel coding.
- **Half-Rate TCH (TCH/H):** Carries speech coded at a lower rate (around 6.5 kbps) or data at up to 4.8 kbps. This allows two users to share a single time slot, effectively doubling the voice capacity of a cell, albeit with some reduction in speech quality. The

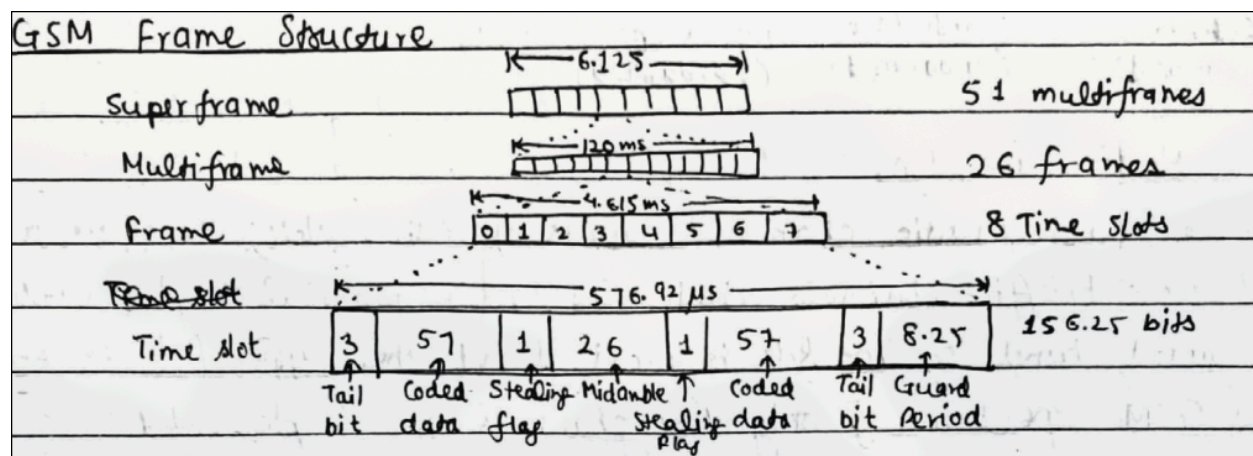
gross data rate is 11.4 kbps.

3.2.2 Control Channels (CCH)

Control Channels are used to carry signaling and synchronization information between the mobile station and the base station.¹ They are further divided into three categories:

1. **Broadcast Channels (BCH) (Downlink only):** These channels are used by the base station to broadcast information to all mobile stations within a cell.
 - **Broadcast Control Channel (BCCH):** Continuously transmits cell-specific information, such as the cell identity, frequency allocations, and network parameters, which a mobile needs to register with the network.
 - **Frequency Correction Channel (FCCH):** Transmits a fixed-frequency burst that allows the mobile station to synchronize its internal frequency to that of the base station.
 - **Synchronization Channel (SCH):** Transmits a burst containing the TDMA frame number and Base Station Identity Code (BSIC), allowing the mobile to synchronize its timing with the cell's frame structure.
2. **Common Control Channels (CCCH):** These channels are used for paging mobile stations and for establishing the initial connection.
 - **Paging Channel (PCH) (Downlink):** Used by the network to alert (page) a mobile station about an incoming call or SMS.
 - **Random Access Channel (RACH) (Uplink):** Used by the mobile station to request a dedicated channel from the network, for example, to originate a call or respond to a page. Multiple mobiles can attempt to access this channel simultaneously, leading to a possibility of collisions.
 - **Access Grant Channel (AGCH) (Downlink):** Used by the network to assign a dedicated channel (an SDCCH) to a mobile station in response to a successful request on the RACH.
3. **Dedicated Control Channels (DCCH):** These channels are assigned to a specific mobile station for signaling purposes during call setup or an active call.
 - **Standalone Dedicated Control Channel (SDCCH):** Used for registration, authentication, location updating, and call setup procedures before a traffic channel is assigned.
 - **Slow Associated Control Channel (SACCH):** Transmitted alongside a TCH or SDCCH. It carries slow but regularly changing control information, such as power control commands and timing advance measurements, in both uplink and downlink directions.
 - **Fast Associated Control Channel (FACCH):** A "pre-emptive" channel used for urgent signaling messages, such as handoff commands. It operates by "stealing" a traffic channel burst, replacing user data with signaling data for one or more frames. A "stealing flag" in the burst indicates whether it contains user data or FACCH data.

3.3 GSM Frame Structure Hierarchy [FIG]



The TDMA scheme in GSM is built upon a complex hierarchical frame structure to organize the flow of data and signaling.¹

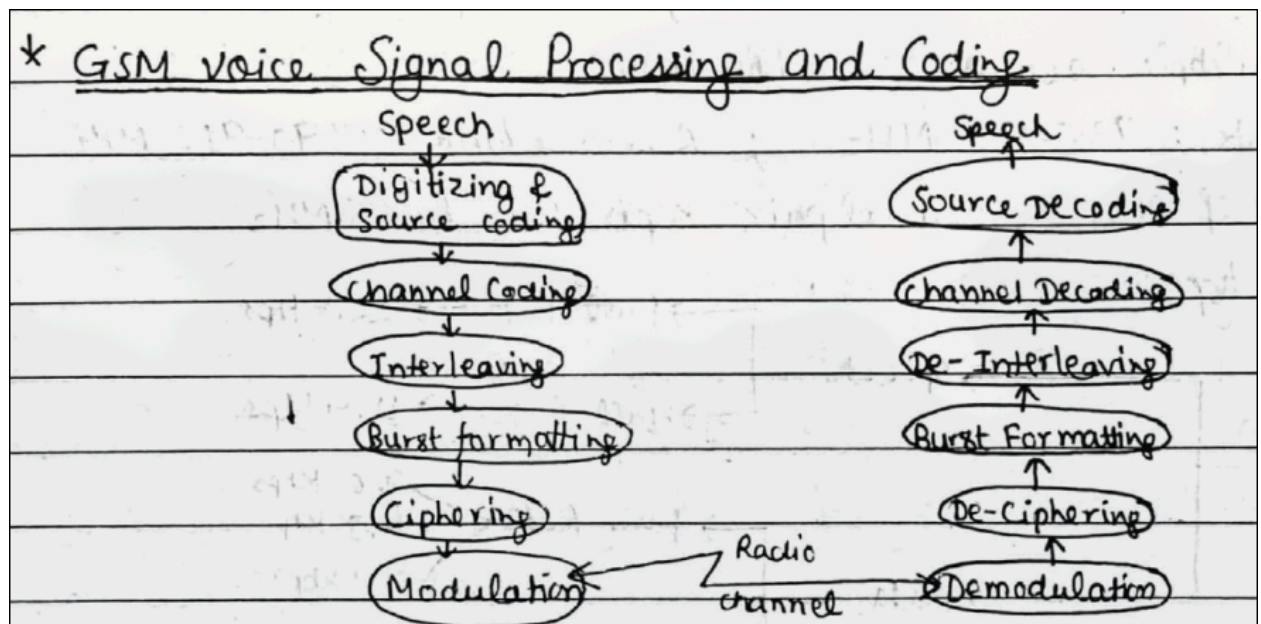
- **Time Slot (Burst):** The smallest unit, lasting 15/26 ms (approx. 576.9 μs) and containing 156.25 bits.
- **TDMA Frame:** A sequence of 8 time slots, numbered 0 to 7. A TDMA frame lasts for 4.615 ms.
- **Multiframe:** A group of TDMA frames. The structure depends on the channel type:
 - A **Traffic Multiframe** consists of 26 TDMA frames and is used for TCH, SACCH, and FACCH. It has a duration of 120 ms.
 - A **Control Multiframe** consists of 51 TDMA frames and is used for CCH and SDCCH. It has a duration of 235.4 ms.
- **Superframe:** Composed of 51 traffic multiframe or 26 control multiframe, for a total of 1326 TDMA frames. A superframe lasts for 6.12 seconds.
- **Hyperframe:** The largest unit, consisting of 2048 superframe (2,715,648 TDMA frames). A hyperframe lasts for approximately 3.5 hours. The hyperframe numbering is used for advanced encryption and frequency hopping sequences.

3.4 Voice and Data Signal Processing Chain [FIG]

To transmit speech or data over the error-prone radio interface, GSM employs a sophisticated multi-step signal processing chain. The process is reversed at the receiving end.¹

1. **Speech Coding:** A 20 ms sample of speech is digitized and then compressed by a speech coder. GSM uses a technique called Residually Excited Linear Predictive Coding with a Long-Term Predictor (RPE-LTP). This coder converts the speech sample into a block of 260 bits, resulting in a net data rate of 13 kbps.
2. **Channel Coding:** This step adds redundant bits to the 260-bit block to protect it from

errors during transmission. The bits are prioritized based on their importance to speech

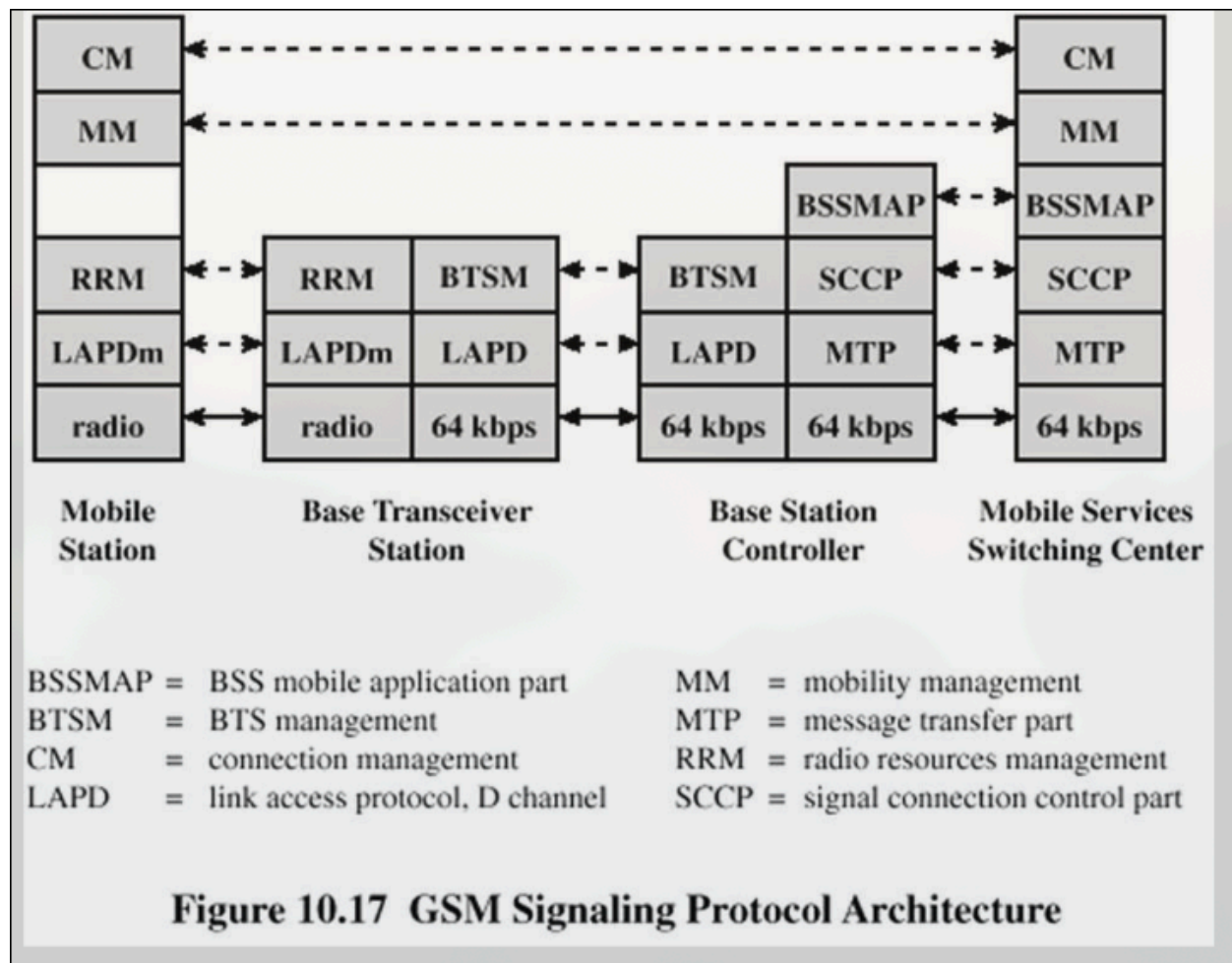


quality. The 50 most critical bits (Type Ia) have a 3-bit Cyclic Redundancy Check (CRC) added for error detection. Then, convolutional coding is applied to all bits, expanding the 260-bit block to a 456-bit block.

3. **Interleaving:** This is a crucial step for combating burst errors caused by fading. The 456 bits from a single speech frame are not transmitted consecutively. Instead, they are reordered and spread across eight consecutive TDMA bursts. This ensures that if one burst is lost due to a deep fade, the errors are distributed across the original speech frame and can be more effectively corrected by the channel decoder.
4. **Ciphering:** For security, the interleaved data is encrypted using the A5 stream cipher algorithm. A unique encryption key (K_c) is generated for each call during the authentication process.
5. **Burst Formatting:** The encrypted bits are assembled into a 148-bit burst (also called a time slot). This involves adding a 26-bit training sequence (midamble) for equalization at the receiver, two "stealing flags" to indicate if the burst is used for FACCH, and tail bits.
6. **Modulation:** The final stream of bits is modulated onto the radio frequency carrier using Gaussian Minimum Shift Keying (GMSK) with a bandwidth-time product (BT) of 0.3. GMSK is a constant envelope modulation scheme, which is power-efficient and well-suited for the amplifiers used in mobile devices.

4.0 GSM Network Signaling, Mobility, and Security

4.1 GSM Signaling Protocol Architecture [FIG]



The communication between the various entities in the GSM network is governed by a structured signaling protocol stack, which is broadly based on the OSI model. The protocol is divided into three layers ¹:

- **Layer 1 (Physical Layer):** Defines the physical transmission medium. On the Um interface, this is the radio link with its TDMA/FDMA structure. On the Abis and A interfaces, it is typically a 64 kbps PCM channel on a wired link.
- **Layer 2 (Data Link Layer):** Ensures reliable data transfer across a single link. The protocol used on the Um interface is Link Access Procedure on the Dm channel (LAPDm), which is optimized for the radio environment. On the A interface, Message Transfer Part (MTP) layers 1-3 of the SS7 protocol suite are used.
- **Layer 3 (Network Layer):** This layer is responsible for the overall control of communication and is divided into three sublayers:
 - **Radio Resource Management (RRM):** Manages the physical and logical radio resources, handling functions like channel assignment, handoff control, and power control.
 - **Mobility Management (MM):** Manages functions arising from user mobility, such

as location updating, registration, and authentication.

- **Connection Management (CM):** Handles the end-to-end setup, maintenance, and termination of calls. It also manages supplementary services and SMS.

4.2 Mobility Management ``

Mobility management is the set of functions that allows the network to track the location of mobile subscribers to route calls to them and maintain connections as they move.¹

4.2.1 Location Updating Procedures

To efficiently route calls, the network needs to know the approximate location of every mobile station. GSM groups cells into larger **Location Areas (LAs)**. A mobile station only needs to inform the network of its location when it moves from one LA to another. This process, known as a Location Update, reduces the signaling overhead compared to updating on every cell change.¹

The process is as follows:

1. The MS periodically listens to the BCCH of the current cell, which broadcasts the Location Area Identity (LAI).
2. When the MS detects that the LAI has changed, it initiates a location update procedure by sending a "Location Update Request" message to the new MSC/VLR via the new BSS.
3. The new VLR communicates with the HLR, informing it of the subscriber's new location.
4. The HLR updates its records with the address of the new VLR and sends an acknowledgement.
5. The HLR also sends a "Cancel Location" message to the old VLR, instructing it to delete the subscriber's now-obsolete record.
6. The new VLR informs the MS that the location update was successful.¹

The design of Location Areas involves a critical trade-off. Small LAs require frequent location updates, increasing signaling traffic. Large LAs reduce this traffic but require the network to page the mobile station across a much wider area for an incoming call, consuming more radio resources on the Paging Channel (PCH).

4.2.2 Handoff (Handover) Scenarios in GSM

Handoff (or handover) is the process of transferring an active call or data session from one channel to another without interruption. In GSM, this is a network-controlled, mobile-assisted process. The MS continuously measures the signal strength of the current cell and neighboring cells and reports these measurements to the BSC. The BSC and MSC use this

information to decide when and where a handoff is needed.¹

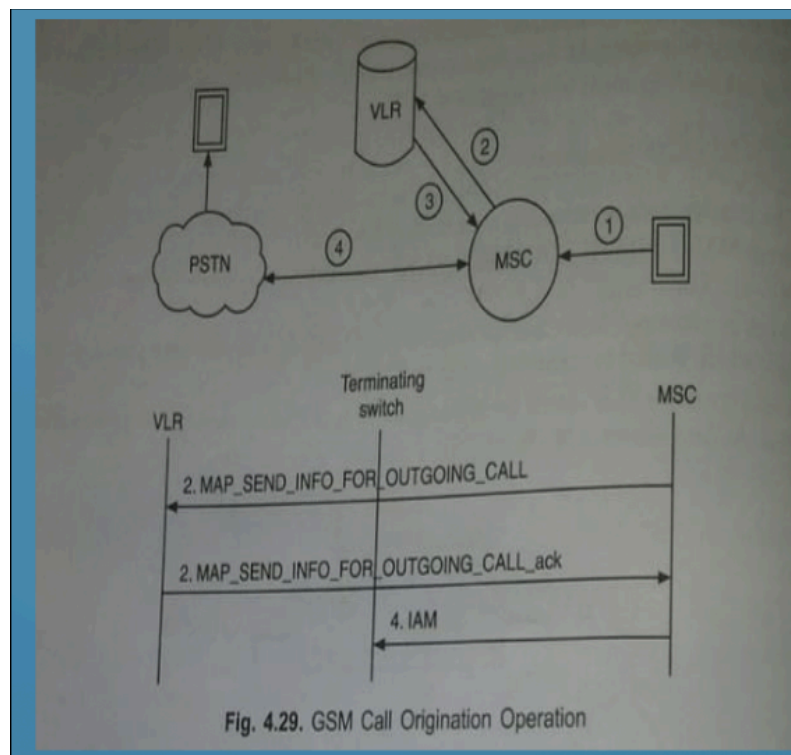
GSM employs a "**hard handoff**" mechanism, which is a "break-before-make" process. The connection to the old channel is released *before* the connection to the new channel is established. This results in a very brief interruption (typically less than 100 ms) in the communication path.¹

There are four main handoff scenarios in GSM:

1. **Intra-cell Handoff:** The MS is transferred to a different channel (frequency or time slot) within the same cell. This may be done to mitigate interference.¹
2. **Inter-cell, Intra-BSC Handoff:** The MS moves from one cell to another, where both cells are controlled by the same BSC. The BSC manages this handoff internally.¹
3. **Inter-BSC, Intra-MSC Handoff:** The MS moves to a cell controlled by a different BSC, but both BSCs are connected to the same MSC. The MSC coordinates the handoff between the two BSCs.¹
4. **Inter-MSC Handoff:** This is the most complex scenario, occurring when the MS moves into a cell controlled by a new MSC. The handoff process must be coordinated between the original MSC and the new target MSC.¹

4.3 Call Routing Procedures

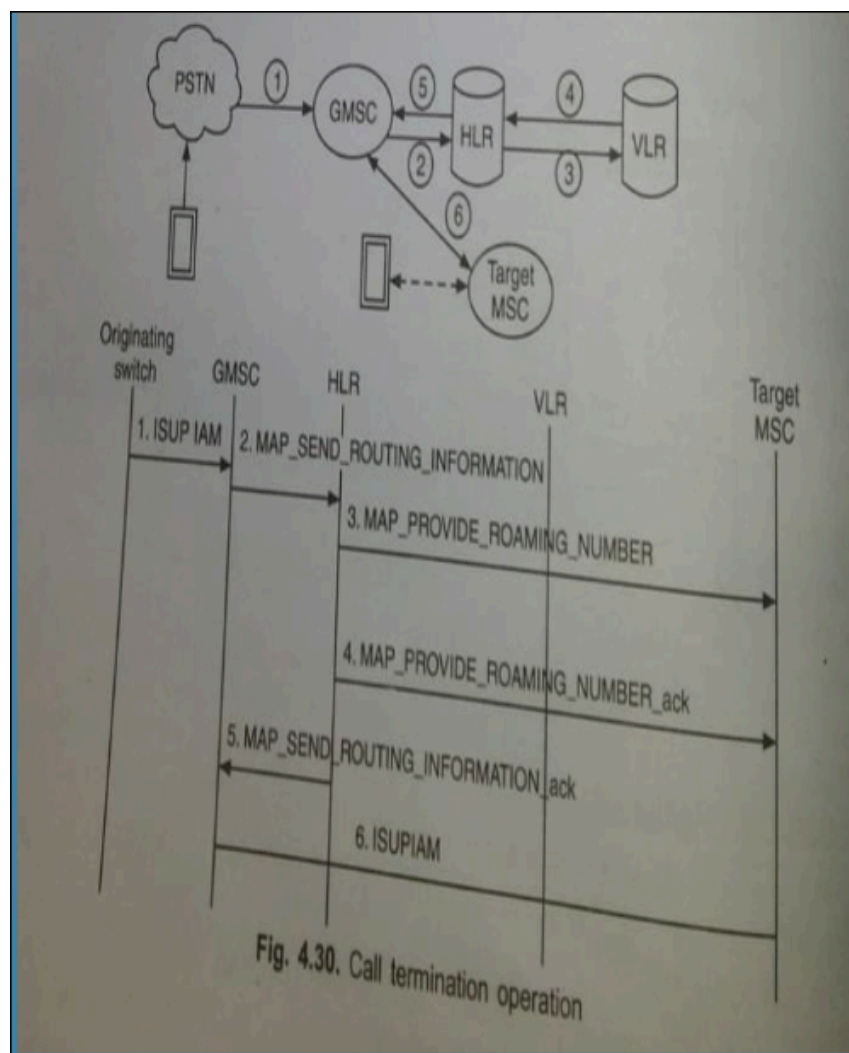
4.3.1 Mobile Call Origination [FIG]



The process for a mobile user to initiate a call is as follows ¹:

1. The MS requests a dedicated channel (SDCCH) from the network using the RACH.
2. The network assigns an SDCCH via the AGCH.
3. The MS sends the dialed number to the MSC over the SDCCH.
4. The MSC forwards the request to its VLR to check if the subscriber is authorized to make the call and has the requested service provisioned.
5. If authorized, the MSC instructs the BSC to allocate a traffic channel (TCH).
6. The MSC then routes the call to the GMSC, which forwards it to the appropriate external network (e.g., PSTN) to reach the called party.¹

4.3.2 Mobile Call Termination [FIG]



Routing a call to a mobile subscriber is more complex because the user's location is not fixed ¹:

1. A call from the PSTN arrives at the GMSC of the subscriber's home network.
2. The GMSC does not know the subscriber's current location, so it queries the subscriber's HLR using their phone number (MSISDN).
3. The HLR checks its database and finds the address of the VLR where the subscriber is currently roaming.
4. The HLR then requests a temporary routing number, the **Mobile Station Roaming Number (MSRN)**, from that VLR.
5. The VLR allocates a temporary MSRN from its pool and sends it back to the HLR.
6. The HLR forwards the MSRN to the GMSC.
7. The GMSC now uses the MSRN to route the call to the correct "visited" MSC, just like a standard telephone call.
8. The visited MSC receives the call, retrieves the subscriber's IMSI from the MSRN, and instructs the BSCs in the subscriber's current Location Area to page the MS on the PCH.
9. The MS responds to the page, and the MSC then establishes the final connection.¹

4.4 Security in GSM

GSM was the first cellular standard to incorporate robust security features as part of its design, primarily to prevent call interception and fraudulent use of the network.¹

4.4.1 Authentication and Encryption Principles

GSM security relies on three key elements:

1. **Authentication:** A "challenge-response" mechanism to verify the subscriber's identity. The network challenges the SIM card to prove it knows the secret key (Ki) without ever transmitting the key itself over the air.¹
2. **Confidentiality:** Encryption of all voice and data traffic transmitted over the radio link to prevent eavesdropping.¹
3. **Anonymity:** Use of a temporary identity (Temporary Mobile Subscriber Identity, TMSI) assigned by the VLR to protect the user's permanent identity (IMSI) from being intercepted over the air.

4.4.2 Role of SIM and Security Algorithms

The security architecture is centered around the SIM card and a set of cryptographic algorithms¹:

- **SIM Card:** The SIM is a tamper-resistant smart card that securely stores the subscriber's 128-bit secret key, known as **Ki**. This key is known only to the SIM and the network's Authentication Center (AuC); it never leaves the SIM.

- **Authentication Process:**
 1. The network's AuC generates a 128-bit random number (**RAND**).
 2. The AuC sends this RAND to the MS.
 3. Inside the SIM, the **A3 algorithm** takes the RAND and the secret key Ki as inputs and produces a 32-bit Signed Response (**SRES**).
 4. The MS sends this SRES back to the network.
 5. Meanwhile, the AuC performs the same calculation. If the SRES from the MS matches the one calculated by the AuC, the subscriber is considered authentic.
- **Encryption Key Generation:**
 1. Simultaneously, the **A8 algorithm** (often combined with A3 in practice) inside the SIM uses the same RAND and Ki to generate a 64-bit session key, known as **Kc**.
 2. This Kc is the key used for encrypting and decrypting data for the duration of the call.
- **Encryption:**
 1. The **A5 algorithm** is a stream cipher used to encrypt the data transmitted between the MS and the BTS. It is initialized with the session key Kc and the current TDMA frame number to generate a keystream that is XORed with the data bursts.

5.0 Interim Standard 136 (IS-136 / D-AMPS)

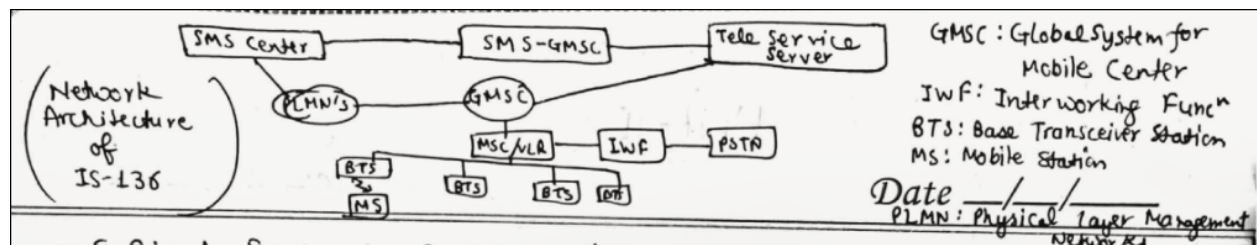
5.1 Overview and Salient Features

Interim Standard 136 (IS-136), also known as Digital AMPS (D-AMPS) or North American Digital Cellular (NADC), was the 2G digital evolution of the 1G Analog Mobile Phone System (AMPS) used primarily in the Americas. It was designed to be backward-compatible, allowing for dual-mode (analog/digital) phones and a gradual upgrade of the existing AMPS network infrastructure.¹

Salient features of IS-136 include:

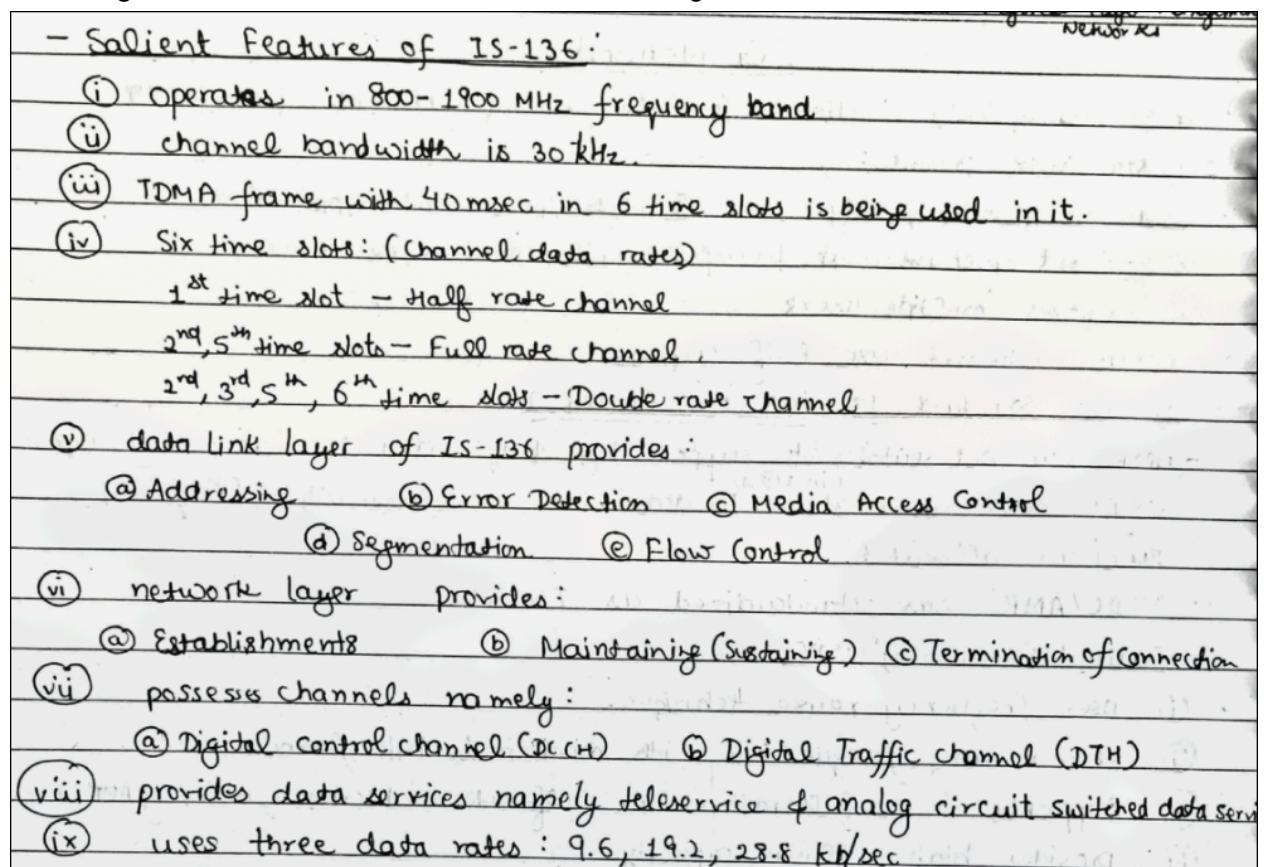
- **Frequency Bands:** Operates in the same 800 MHz cellular band as AMPS and also in the 1900 MHz PCS band.¹
- **Channel Bandwidth:** Retains the 30 kHz channel spacing of AMPS to facilitate coexistence.¹
- **Multiple Access:** Uses a TDMA/FDD scheme. Each 30 kHz radio channel is divided into 6 time slots within a 40 ms frame. This initially provided three full-rate user channels per radio carrier, a 3x capacity increase over AMPS.¹
- **Subscriber Rates:** Supports full-rate, half-rate, and double-rate channels by allocating different numbers of time slots to a user.¹

5.2 IS-136 Architecture and Channels [FIG]

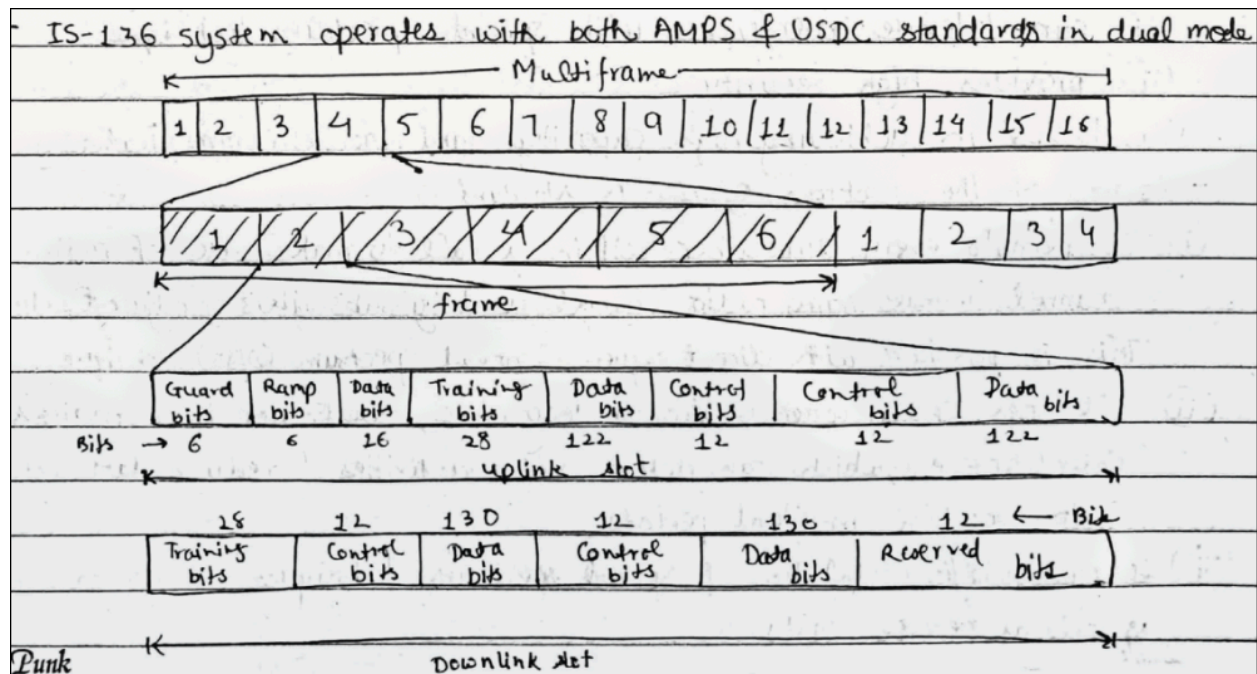


The network architecture of IS-136 is conceptually similar to GSM, consisting of mobile stations, base stations, and a switching center. However, the terminology and specific protocols differ. The two primary logical channel types are ¹:

- **Digital Control Channel (DCCH):** Carries signaling information for call setup, paging, and system parameter broadcasts.
- **Digital Traffic Channel (DTC):** Carries the digitized voice or user data.



5.3 IS-136 TDMA Frame Structure [FIG]



The IS-136 TDMA frame has a duration of 40 ms and is composed of 6 time slots. Each time slot contains fields for data, a training sequence for equalization, and control bits. The structure differs for the uplink and downlink to accommodate the different timing requirements of the mobile and base station transmitters.¹

- Salient Features of USDC	
(i)	uses frequency reuse technique.
(ii)	shares same frequencies with minimized interference.
(iii)	supports three full-rate or six half-rate subscribers on every AMPS channel.
(iv)	provides high cellular capacity.
(v)	accommodates many users.
(vi)	also called North American Digital Cellular Standard (NADC).

6.0 Interim Standard 95 (IS-95 / cdmaOne)

6.1 Core Principles and Advantages

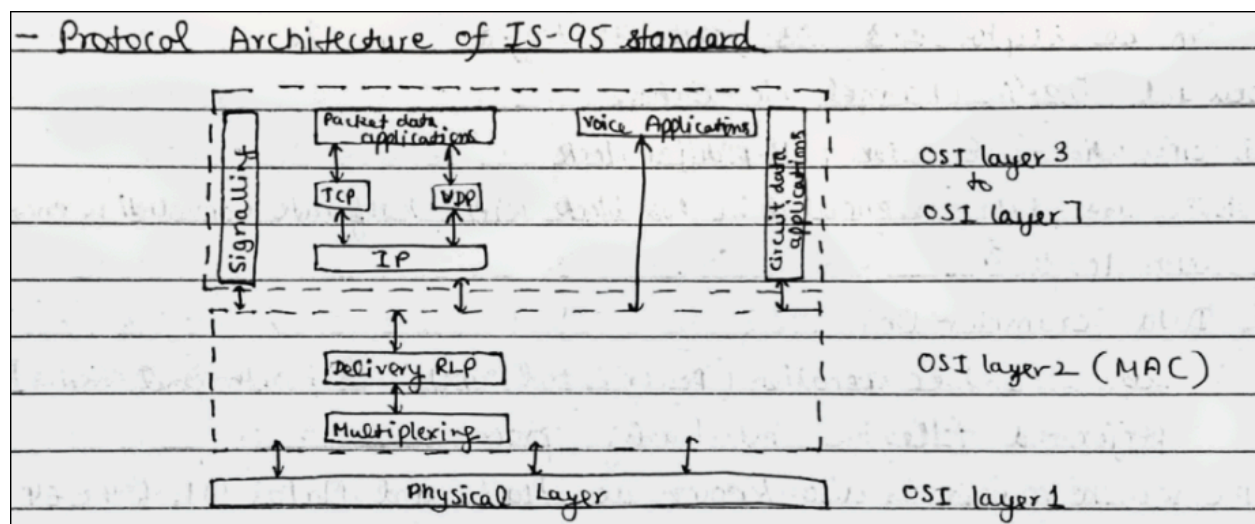
Interim Standard 95 (IS-95), commercially known as cdmaOne, was the first major 2G

standard based on Code Division Multiple Access. Developed by Qualcomm, it represented a significant departure from the TDMA-based approach of GSM and IS-136.¹

The core advantages of its CDMA-based design are:

- **High Capacity:** By allowing all users to share the same frequency, and through features like soft handoff and efficient voice activity detection, IS-95 systems could achieve significantly higher user capacity than TDMA systems in the same amount of spectrum.¹
- **Interference Tolerance:** The spread spectrum technique makes the signal inherently resistant to narrowband interference and multipath fading.¹
- **High Security:** The use of unique spreading codes for each user provides a high degree of privacy and makes the signal difficult to intercept or jam.¹
- **Soft Handoff:** IS-95 supports a "make-before-break" handoff, where a mobile station can be simultaneously connected to two or more base stations, leading to a seamless transition with no dropped calls at cell boundaries.

6.2 IS-95 Protocol Architecture and Channels



In IS-95, channels are not defined by frequency or time but by their unique spreading codes. The system uses a 1.2288 Mcps (mega-chips per second) Direct Sequence Spread Spectrum (DSSS) signal within a 1.25 MHz wide radio channel.¹

6.2.1 Forward Link Channels (Downlink)

The forward link (base station to mobile) consists of up to 64 logical channels that are all transmitted on the same frequency. They are made perfectly orthogonal to each other using a set of 64 **Walsh codes**. This ensures that a mobile station listening for one channel (e.g., its traffic channel) does not receive interference from the other 63 channels transmitted by the same base station.¹

- **Pilot Channel (Walsh Code 0):** An unmodulated, continuous stream of all zeros. It is used by all mobiles in the cell for initial synchronization, timing acquisition, and as a signal strength reference for handoff decisions.
- **Sync Channel (Walsh Code 32):** Provides the mobile with critical system identification and timing information.
- **Paging Channels (Walsh Codes 1-7):** Used to page the mobile for incoming calls and to transmit system overhead information.
- **Forward Traffic Channels (Remaining Walsh Codes):** Used to carry user voice and data to specific mobiles.

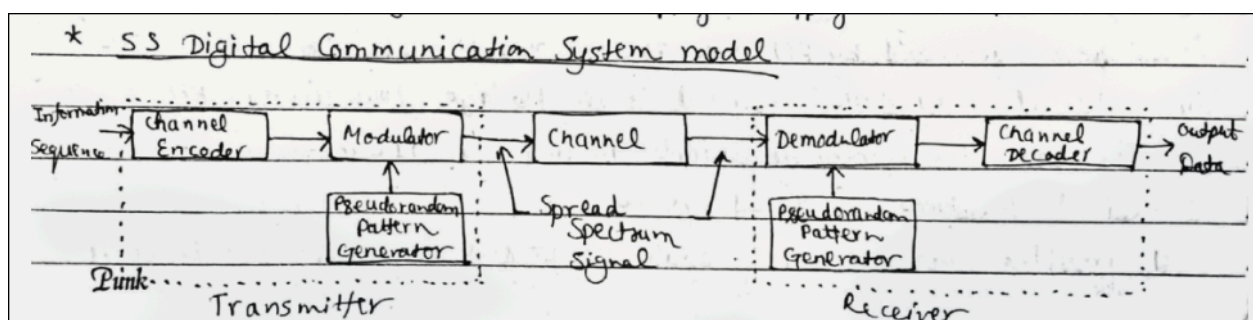
6.2.2 Reverse Link Channels (Uplink)

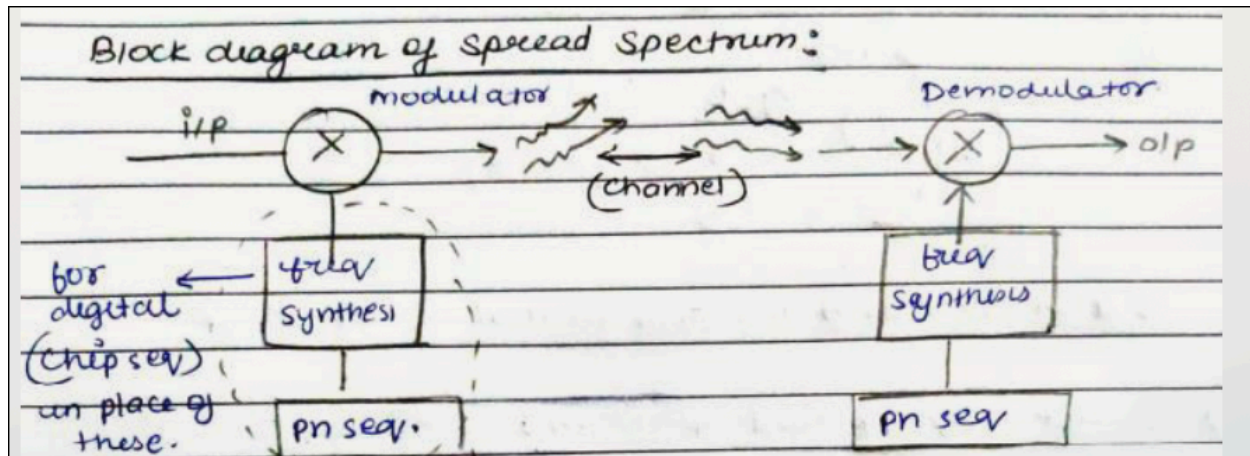
On the reverse link (mobile to base station), it is impossible to use orthogonal Walsh codes because the signals from different mobiles arrive at the base station unsynchronized. Instead, all reverse link channels are separated by using different time offsets of a very long pseudo-random noise (PN) code. The base station can distinguish the signals by correlating the received composite signal with the appropriately time-shifted PN code for each user.¹

- **Access Channels:** Used by the mobile to initiate communication with the base station (e.g., to originate a call or respond to a page).
- **Reverse Traffic Channels:** Used to carry user voice and data from the mobile to the base station.

7.0 Spread Spectrum and CDMA Principles

7.1 Introduction to Spread Spectrum Modulation [FIG]





Spread Spectrum (SS) is a modulation technique where the bandwidth of the transmitted signal, BSS, is deliberately made much larger than the minimum bandwidth required to transmit the information, B. This "spreading" is achieved by multiplying the information signal with a high-rate pseudo-random noise (PN) sequence, also known as a spreading code.¹

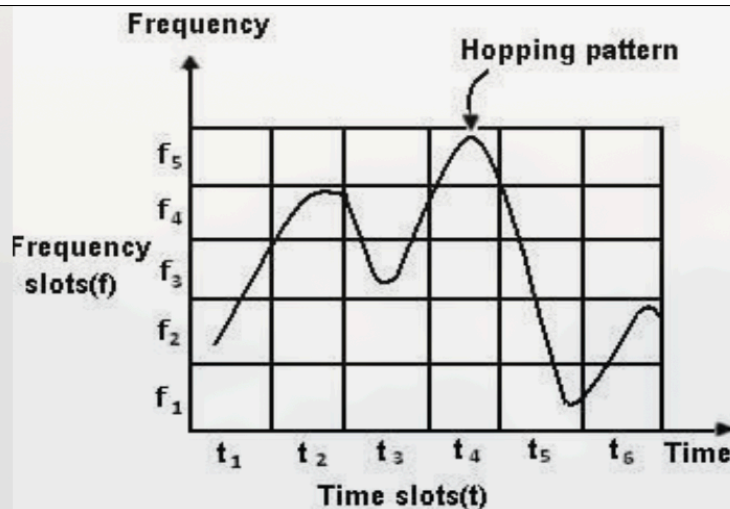
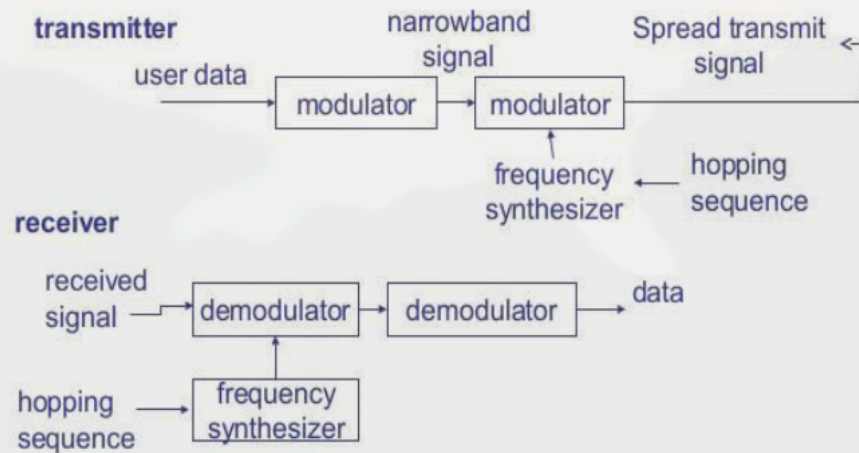
The key characteristics of a spread spectrum signal are:

- The occupied bandwidth is much wider than the information rate.
- The signal energy is spread over this wide bandwidth, making its power spectral density very low, often below the thermal noise floor. This makes the signal difficult to detect, a property known as Low Probability of Intercept (LPI).
- The receiver uses a synchronized copy of the same PN sequence to "de-spread" the signal, collapsing its energy back into the original narrow bandwidth while simultaneously spreading the energy of any narrowband interference, which can then be filtered out.¹

This technique was developed to solve critical communication problems, particularly in military applications, such as combating intentional interference (jamming) and providing secure communication.¹

7.2 Frequency Hopping Spread Spectrum (FH-SS) [FIG]

FHSS (Frequency Hopping Spread Spectrum)



In Frequency Hopping Spread Spectrum (FH-SS), the spectrum is spread sequentially rather than instantaneously. The available wideband is divided into many smaller frequency channels. The transmitter rapidly changes, or "hops," its carrier frequency among these channels in a pseudo-random sequence known only to the transmitter and intended receiver. An unintended listener would only hear random blips of energy across the band.¹

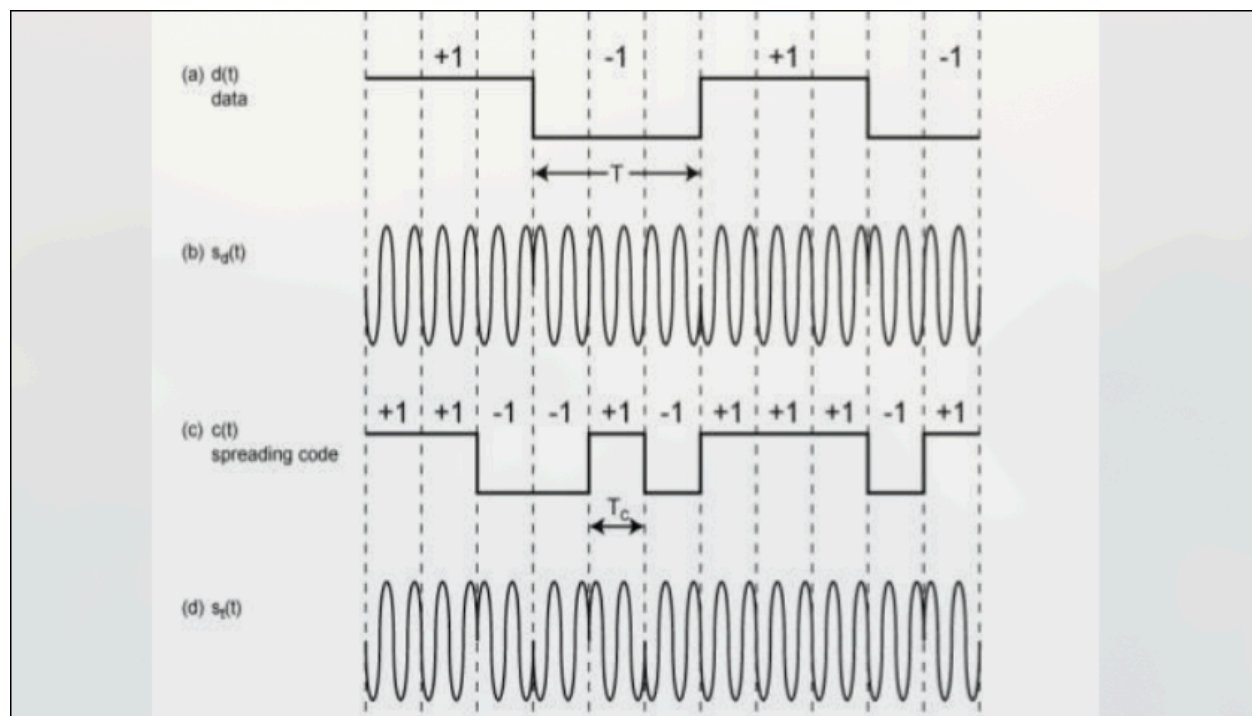
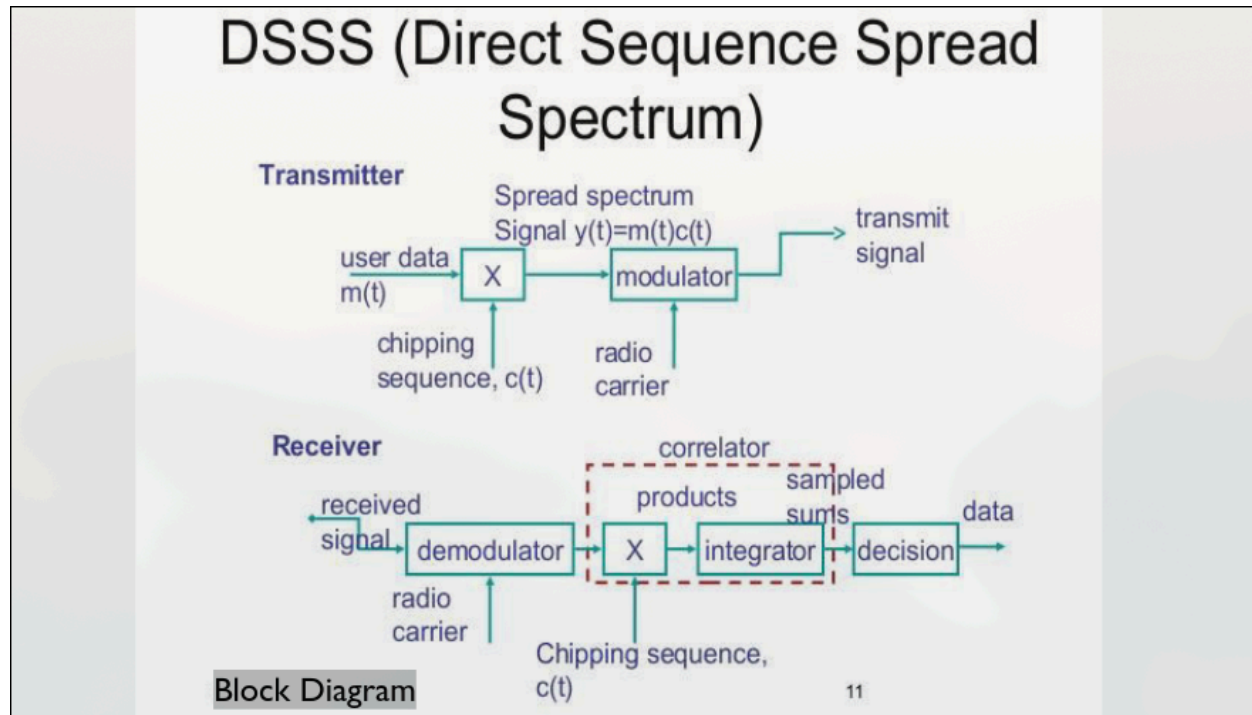
There are two main types of FH-SS:

- **Slow Frequency Hopping:** The transmitter stays on a single frequency for the duration of multiple data symbols. The symbol rate (R_s) is an integer multiple of the hop rate (R_h).
- **Fast Frequency Hopping:** The carrier frequency hops multiple times during the

transmission of a single data symbol. The hop rate (R_h) is an integer multiple of the symbol rate (R_s).

FH-SS is notably used in the Bluetooth wireless personal area network (WPAN) standard.¹

7.3 Direct Sequence Spread Spectrum (DS-SS)



In Direct Sequence Spread Spectrum (DS-SS), the spreading is achieved by directly multiplying the narrowband information signal with a high-rate PN sequence. Each bit of the PN sequence is called a "chip," and the rate of the sequence is the "chip rate." The ratio of the chip rate to the information bit rate is known as the **processing gain**. This process spreads the signal's spectrum by a factor equal to the processing gain. DS-SS is the foundational technology for CDMA systems like IS-95.¹

7.4 Principles of Code Division Multiple Access (CDMA) ``

CDMA is a multiple access technique that applies DS-SS principles to allow multiple users to share the same physical channel (frequency and time) simultaneously. The core idea is to assign a unique spreading code (or signature sequence) to each user.¹

The process works as follows:

1. **Transmission:** Each user's data signal is multiplied by their unique high-rate PN code. This spreads each user's signal over the entire available bandwidth. The signals from all users are then transmitted over the same frequency band at the same time.
2. **Reception:** At the receiver (e.g., the base station), the composite signal is a sum of all the transmitted user signals plus noise. To recover the data for a specific user (e.g., User A), the receiver multiplies the entire composite signal by User A's unique code.
3. **Correlation:** Due to the mathematical properties of the PN codes, when the composite signal is multiplied by User A's code, User A's signal is de-spread back to its original narrow bandwidth, concentrating its energy. The signals from all other users, when multiplied by User A's code, remain as wideband, low-power noise. A simple low-pass filter can then remove this noise, leaving only the desired signal from User A.¹

7.5 Power Control and the Near-Far Problem ``

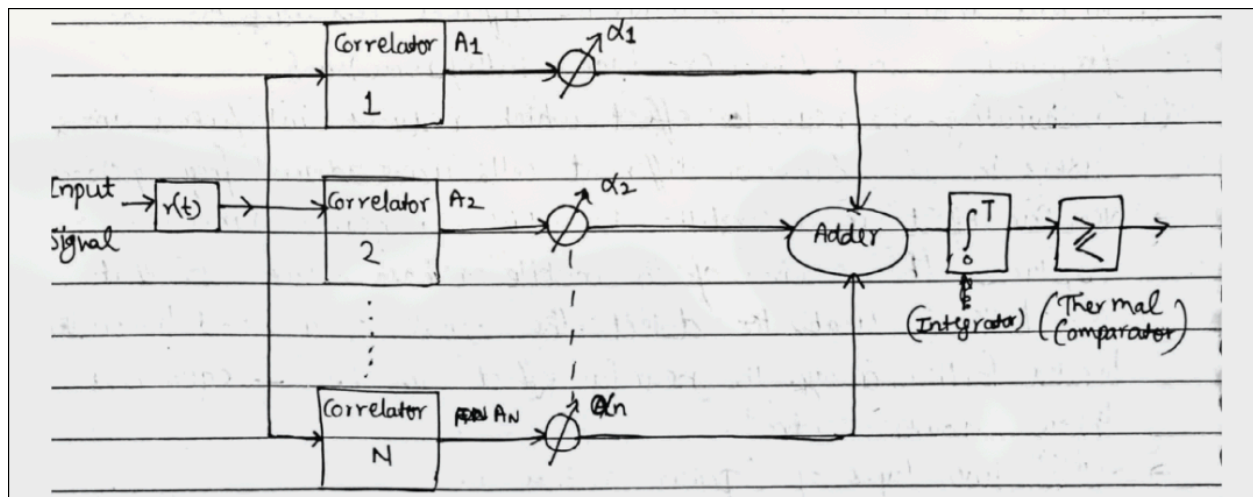
A critical challenge in CDMA systems is the **near-far problem**. Since all users transmit on the same frequency, a mobile station close to the base station will have its signal arrive with much higher power than a mobile station located far away. Without any correction, the strong signal from the nearby user would completely overwhelm the weak signal from the distant user, making it impossible for the base station to decode the distant user's data.¹

To solve this, CDMA systems rely on stringent and rapid **power control**. The goal is to ensure that the signals from all mobile stations arrive at the base station with approximately equal power, regardless of their distance or channel conditions. There are two main types of power control ¹:

- **Open Loop Power Control:** The mobile station estimates the path loss on the downlink (from the base station) and adjusts its own transmit power accordingly, assuming the uplink path loss is similar. This is a fast but less accurate method.

- **Closed Loop Power Control:** The base station measures the received power from the mobile station. If the power is too high or too low, it sends a power-up or power-down command back to the mobile on a dedicated control channel. This feedback loop is more accurate but has a slight delay. In IS-95, these commands are sent 800 times per second.

7.6 Mitigating Multipath Effects with the RAKE Receiver [FIG]



In wireless environments, signals travel along multiple paths from the transmitter to the receiver, arriving at different times. This is known as multipath propagation. In narrowband systems like GSM, this causes inter-symbol interference, which degrades performance and must be corrected by an equalizer.

In wideband CDMA systems, the high chip rate means that the multipath arrivals can often be resolved as distinct, time-delayed versions of the original signal. CDMA systems cleverly turn this problem into an advantage using a **RAKE receiver**.¹

A RAKE receiver consists of multiple parallel correlators, often called "fingers." Each finger can be assigned to lock onto and decode a single strong multipath component. The outputs of all the fingers are then combined in a weighted manner to produce a single, stronger signal. This process is a form of time diversity, as it combines the energy from signals that have traveled different paths, making the overall reception more robust against fading.¹ This ability to exploit multipath is a fundamental advantage of CDMA over narrowband TDMA systems.

8.0 Comparative Analysis of Multiple Access Technologies

The choice of multiple access technology is a fundamental design decision that dictates many of a wireless system's characteristics, including its capacity, complexity, and performance.

The following table provides a detailed comparison of FDMA, TDMA, and CDMA.¹

8.1 Table: FDMA vs. TDMA vs. CDMA

Feature	FDMA (Frequency Division Multiple Access)	TDMA (Time Division Multiple Access)	CDMA (Code Division Multiple Access)
Principle	Bandwidth is divided and shared among many stations.	Time is divided and shared among many stations on the same frequency.	Bandwidth and time are shared simultaneously by all stations.
Resource Allocation	Assigns a unique, non-overlapping frequency band to each user.	Assigns a unique, recurring time slot to each user.	Assigns a unique pseudo-random code to each user.
Synchronization	Not necessary between users.	Essential for precise time slot alignment to prevent crosstalk.	Essential for code alignment at the receiver for successful de-spreading.
Guard Bands/Times	Requires guard bands between adjacent frequency channels to prevent interference.	Requires guard times between adjacent time slots to absorb timing errors.	Does not require hard guard bands/times; separation is achieved through code orthogonality.
Interference	Prone to intermodulation distortion and adjacent channel interference.	Prone to interference between adjacent time slots (crosstalk) if timing is off.	All users act as co-channel interference to each other. System performance is interference-limited.
Capacity	Hard Capacity: Fixed number of channels. System is blocked when all channels are full.	Hard Capacity: Fixed number of time slots. System is blocked when all slots are full.	Soft Capacity: No fixed user limit. As more users are added, interference increases for everyone, gradually degrading the quality of all calls. The capacity is limited by the acceptable level of interference.
Handoff	Hard Handoff	Hard Handoff	Soft Handoff

	(break-before-make).	(break-before-make).	(make-before-break), allowing simultaneous connection to multiple base stations.
Example Systems	1G Analog Systems (e.g., AMPS).	2G Digital Systems (e.g., GSM, IS-136).	2G/3G Digital Systems (e.g., IS-95, W-CDMA, CDMA2000).